

Euler's Theorem

Definition:

Euler's Theorem (for Undirected Graphs):

A connected graph has an Eulerian circuit (i.e., a circuit that uses every edge exactly once and starts and ends at the same vertex) if and only if all vertices have even degree.

A connected graph has an Eulerian trail (uses every edge exactly once, but starts and ends at different vertices) if and only if exactly two vertices have odd degree and all others have even degree.

Terms:

Eulerian Trail: A trail that visits every edge exactly once, starting and ending at different vertices.

Eulerian Circuit: A trail that visits every edge exactly once, and starts and ends at the same vertex.

Degree of a vertex: Number of edges incident to the vertex.

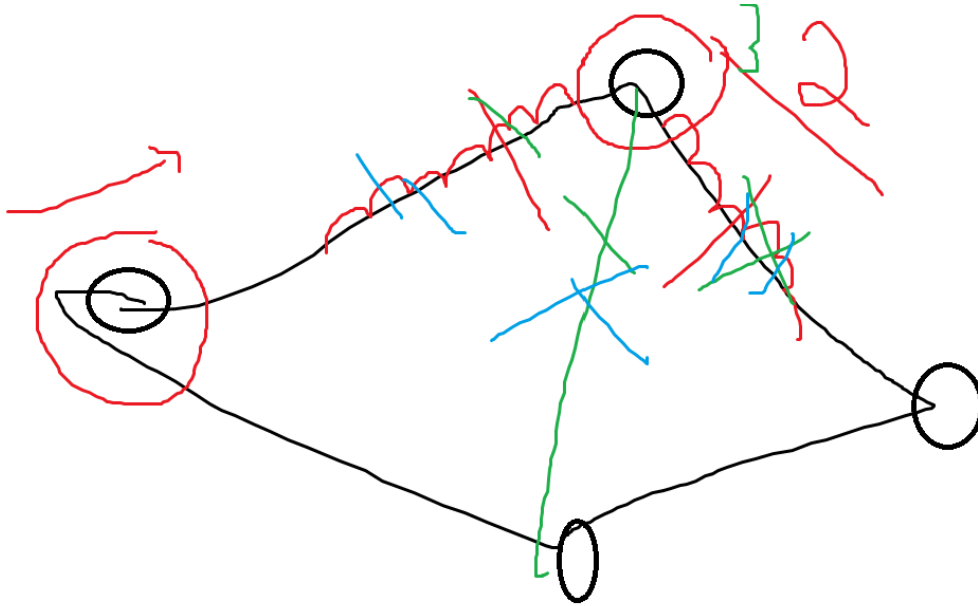
Connected graph: Every vertex can be reached from any other vertex

Formula:

Let $G=(V,E)$ be a finite, connected undirected graph.

1. Eulerian Circuit exists $\Leftrightarrow$ all vertices have even degree.
2. Eulerian Trail exists $\Leftrightarrow$ exactly two vertices have odd degree.
3. No Euler path/circuit $\Leftrightarrow$ more than two vertices have odd degree.

Proof:



Part 1: If G has an Eulerian circuit, then all vertices have even degree

Let G be connected and contain an Eulerian circuit.

- Every time we enter a vertex v , we must leave it (since we are in a circuit).
- Thus, the edges incident on v come in pairs — one edge in, one edge out.
- Therefore, the degree of each vertex is even, since each traversal contributes 2 to the degree (in + out).

Hence, all vertices have even degree.

Part 2: If all vertices have even degree and G is connected, then it has an Eulerian circuit

We will prove this using induction and constructive approach (Fleury's Algorithm-like logic):

Construction Approach / Inductive Proof

1. Base Case:

If the graph has 0 edges, then the circuit is trivially the starting vertex (no edges needed). The theorem holds.

2. Inductive Step:

- Start at any vertex v .
- Since all degrees are even, you can always exit a vertex if you enter it (no dead ends).
- Use Depth First Search (DFS) or Fleury's idea: always move to a non-bridge edge unless no other choice.
- Continue until a closed circuit is formed, i.e., back to starting vertex.

3. If not all edges used:

- Then remaining edges form a subgraph (also with even degrees).
- This subgraph is connected (or has connected components with even degrees).
- Apply the same logic recursively to form another circuit.
- Splice this new circuit into the original.

4. Eventually, you will have a single circuit that uses every edge exactly once — an Eulerian circuit.

Hence, the graph has an Eulerian circuit.

Applications of Euler's Theorem

- Route planning (e.g., postal delivery, garbage collection)
- Computer graphics (tracing outlines)
- Puzzles like Königsberg bridge problem

- Network analysis